

Intro to Python

Course so far

So far, we know how to do key wrangling tasks in R:

- Grouping, summarizing, mutating
- Pivoting and joining

What is next: Data wrangling in Python!

What *is* R?

- R is a programming language specifically designed for statistics and data analysis
 - Objects for storing data, and functions for interacting with data, are fundamental
 - R is very good at graphics and visualization
 - R is easily extended. Users can write and share their own functions and packages
- We can interact with R through IDEs like RStudio

What other options exist?

- SAS
- Stata
- SPSS
- Excel
- Python
- Julia
- Matlab
- Many others...

What is Python

- Python is a general-purpose programming language
- Like R, python has a wide range of packages to extend functionality
- Certain Python packages allow for sophisticated data analysis and modeling
 - SciPy, NumPy
 - scikit-learn, statsmodels, pytorch
 - pandas
 - matplotlib

R vs. Python

My own, *personal*, preferences:

R is good for

- Data visualization and wrangling
- Classical statistics
- Statistical inference
- New statistical methods

Python is good for

- General-purpose programming
- Challenging data types (e.g. images)
- Prediction and machine learning

Warmup

What do you think this code is doing? Discuss with your neighbor for 1-2 minutes, then we will discuss as a class.

```
1 import pandas as pd
2 import seaborn as sns
3
4 penguins = sns.load_dataset("penguins")
5
6 (penguins.groupby('species').
7   agg(
8     mean_bill = ('bill_length_mm', 'mean'),
9     sd_bill = ('bill_length_mm', 'std')
10  ))
```

R: library (...)

penguins |>

group_by (species) |>

summarize(mean_bill = mean(bill_length_mm), ...)

Warmup

package for working with data

short alias for pandas package

```
1 import pandas as pd
2 import seaborn as sns
3
4 penguins = sns.load_dataset("penguins")
5
6 (penguins.groupby('species').
7  agg(
8      mean_bill = ('bill_length_mm', 'mean'),
9      sd_bill = ('bill_length_mm', 'std')
10 ))
```

pairing column with
the function to apply

need quotes around
column names in
pandas

	mean_bill	sd_bill
species		
Adelie	38.791391	2.663405
Chinstrap	48.833824	3.339256
Gentoo	47.504878	3.081857

Data wrangling ideas

- choose certain rows and columns
- calculate summary statistics
- group rows together
- create new columns
- apply functions across columns
- reshape data by pivoting
- join tables

These ideas are language-agnostic! The implementation is just a bit different

Basic information

```
1 penguins
```

```
      species      island bill_length_mm ... flipper_length_mm
body_mass_g      sex
0    Adelie  Torgersen      39.1 ...      181.0
3750.0    Male
1    Adelie  Torgersen      39.5 ...      186.0
3800.0  Female
2    Adelie  Torgersen      40.3 ...      195.0
3250.0  Female
3    Adelie  Torgersen      NaN ...      NaN
NaN      NaN
4    Adelie  Torgersen      36.7 ...      193.0
3450.0  Female
..      ...      ...      ...      ...
```

Basic information

```
1 penguins.shape
```

```
In R: dim(penguins)
```

```
(344, 7)
```

```
1 penguins.columns
```

```
In R: colnames(penguins)
```

```
Index(['species', 'island', 'bill_length_mm', 'bill_depth_mm',  
      'flipper_length_mm', 'body_mass_g', 'sex'],  
      dtype='object')
```

Choosing columns

```
1 penguins['bill_length_mm']
```

0 39.1

1 39.5

2 40.3

3 NaN

4 36.7

...

339 NaN

340 46.8

341 50.4

342 45.2

343 49.9

Name: bill_length_mm, Length: 344, dtype: float64

In R:

`penguins $ bill-length-mm`

`penguins[["bill-length-mm"]]`

`penguins |>`

`select(bill-length-mm)`

Choosing columns

```
1 penguins[['bill_length_mm', 'bill_depth_mm']]
```

	bill_length_mm	bill_depth_mm
0	39.1	18.7
1	39.5	17.4
2	40.3	18.0
3	NaN	NaN
4	36.7	19.3
...	...	...
339	NaN	NaN
340	46.8	14.3
341	50.4	15.7
342	45.2	14.8
343	49.9	16.1

`['bill-length-mm', 'bill-depth-mm']`

is a list of column
names

Calculating summary statistics

```
1 penguins.agg({'bill_length_mm': 'mean'})
```

```
bill_length_mm    43.92193  
dtype: float64
```

```
1 penguins.agg(mean_length = ('bill_length_mm', 'mean'))
```

```
              bill_length_mm  
mean_length    43.92193
```

Summarizing multiple columns

```
1 penguins[['bill_length_mm', 'bill_depth_mm']].agg('mean')
```

```
bill_length_mm    43.92193
bill_depth_mm     17.15117
dtype: float64
```

```
1 penguins[['bill_length_mm', 'bill_depth_mm']].agg(['mean', 'std'])
```

	bill_length_mm	bill_depth_mm
mean	43.921930	17.151170
std	5.459584	1.974793

Grouping and summarizing

```
1 (penguins.groupby('species').  
2   agg(  
3     mean_bill = ('bill_length_mm', 'mean'),  
4     sd_bill = ('bill_length_mm', 'std')  
5   )) ← need parens to break up expression over multiple lines
```

	mean_bill	sd_bill
species		
Adelie	38.791391	2.663405
Chinstrap	48.833824	3.339256
Gentoo	47.504878	3.081857

Grouping and summarizing

```
1 penguins |>  
2   group_by(island) |>  
3   summarize(mean_length = mean(bill_length_mm))
```

A tibble: 3 × 2

	island	mean_length
	<fct>	<dbl>
1	Biscoe	NA
2	Dream	44.2
3	Torgersen	NA

Grouping and summarizing

```
1 penguins |>
2   group_by(species) |>
3   summarize(mean_length = mean(bill_length_mm,
4                                 na.rm = T))
```

A tibble: 3 × 2

	species <fct>	mean_length <dbl>
1	Adelie	38.8
2	Chinstrap	48.8
3	Gentoo	47.5

pandas automatically skips
missing values!

Sorting

```
1 (penguins.groupby('species').  
2   agg(  
3     mean_bill = ('bill_length_mm', 'mean'),  
4     sd_bill = ('bill_length_mm', 'std')  
5   ).  
6   sort_values('mean_bill'))
```

	mean_bill	sd_bill
species		
Adelie	38.791391	2.663405
Gentoo	47.504878	3.081857
Chinstrap	48.833824	3.339256

Sorting

```
1 (penguins.groupby('species').  
2   agg(  
3     mean_bill = ('bill_length_mm', 'mean'),  
4     sd_bill = ('bill_length_mm', 'std')  
5   ).  
6   sort_values('mean_bill', ascending = False))
```

	mean_bill	sd_bill
species		
Chinstrap	48.833824	3.339256
Gentoo	47.504878	3.081857
Adelie	38.791391	2.663405

Class activity

- Work with a neighbor on the class activity
- You do not need to submit this activity!